



Full length article

1,2-Dichloroethane induces testicular pyroptosis by activating piR-mmu-1019957/IRF7 pathway and the protective effects of melatonin

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ABSTRACT

1,2-Dichloroethane (1,2-DCE) is a prevalent environmental contaminant, and our study revealed its induction of testicular toxicity in mice upon subacute exposure. Melatonin, a prominent secretory product of the pineal gland, has been shown to offer protection against pyroptosis in male reproductive toxicity. However, the exact mechanism underlying 1,2-DCE-induced testicular toxicity and the comprehensive extent of melatonin's protective effects in this regard remain largely unexplored. Therefore, we sequenced testis piRNAs in mice exposed to environmentally relevant concentrations of 1,2-DCE by 28-day dynamic inhalation, and investigated the role of key piRNAs using GC-2 spd cells. Our results showed that 1,2-DCE induced mouse testicular damage and GC-2 spd cell pyroptosis. 1,2-DCE upregulated the expression of pyroptosis-correlated proteins in both mouse testes and GC-2 spd cells. 1,2-DCE exposure caused pore formation on cellular membranes and lactate dehydrogenase leakage in GC-2 spd cells. Additionally, we identified three upregulated piRNAs in 1,2-DCE-exposed mouse testes, among which piR-mmu-1019957 induced pyroptosis in GC-2 spd cells, and its inhibition alleviated 1,2-DCE-induced pyroptosis. piR-mmu-1019957 mimic and 1,2-DCE treatment activated the expression of interferon regulatory factor 7 (IRF7) in GC-2 spd cells. IRF7 knockdown reversed 1,2-DCE-induced cellular pyroptosis, and overexpression of piR-mmu-1019957 did not promote pyroptosis when IRF7 was inhibited. Notably, melatonin reversed 1,2-DCE-caused testicular toxicity, cellular pyroptosis, and upregulated piR-mmu-1019957 and IRF7. Collectively, our findings indicated that melatonin mitigates this effect, suggesting its potential as a therapeutic intervention against 1,2-DCE-induced male reproductive toxicity in clinical practice.

1. Introduction

Synthetic halogenated hydrocarbon 1,2-dichloroethane (1,2-DCE, CAS 107-06-2) is widely used in the manufacture of polyvinyl chloride, and serves as an industrial solvent in China and around the world

(Janssen et al., 1994). Unfortunately, its ubiquitous use in industries and agriculture has resulted in frequent environmental contamination through industrial spills and unauthorized emissions (Olaniran et al., 2009). Consequently, 1,2-DCE is detected in various environmental compartments, including the atmosphere, groundwater, surface water,

Abbreviations: 1,2-DCE, 1,2-dichloroethane; ANOVA, analysis of variance; CCK-8, cell counting kit-8; DMEM, Dulbecco's modified Eagle's medium; DMSO, dimethyl sulfoxide; FBS, fetal bovine serum; FC, fold change; FPKM, fragments per kilobase of transcript per million mapped fragments; GC-MS, gas chromatography-mass spectrometry; H&E, hematoxylin-eosin; KEGG, Kyoto encyclopedia of genes and genomes; LDH, lactate dehydrogenase; PBS, phosphate-buffered saline; NIH, National Institutes of Health; PI, propidium iodide; PIWI, P-element-induced wimpy; piRNA, p-element-induced wimpy-interacting RNAs; qPCR, quantitative polymerase chain reactions; SD, standard deviation; SEM, scanning electron microscopy; siRNAs, small interfering RNAs; TWA, time-weighted average.

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and even drinking water (ATSDR, 2001; IARC, 1979; Xie et al., 2021). The U.S. Environmental Protection Agency estimates that 0.3 % of all groundwater supplies contain 1,2-DCE concentrations ranging from 0.5 to 5.0 µg/L (USEPA, 1987). A recent report has raised concerns about the presence of alarming concentrations of 1,2-DCE in groundwater contaminated by industrial activities, with levels reaching as high as 1.68×10^7 µg/L (Tang et al., 2022). Furthermore, inhalation of high levels of 1,2-DCE (500–1500 mg/m³) in poorly ventilated indoor environments has been associated with encephalopathy in humans (Chen et al., 2015). Apart from industrial sources, the use of substandard indoor decoration materials, the influx of outdoor air pollution, and the burning of coal in rural areas contribute to the elevated levels of 1,2-DCE in residential air samples (Chin et al., 2014; Huang et al., 2019; Xie et al., 2021). Human exposure to 1,2-DCE primarily occurs through inhalation of contaminated air and ingestion of contaminated drinking water (Dai et al., 2017; Hughes et al., 1994).

Given the increasing ubiquity of 1,2-DCE as an environmental pollutant, its toxic effects are garnering growing attention (Chen et al., 2021; Huang et al., 2022). 1,2-DCE has been shown to induce toxicity in the central nervous system, liver, kidneys, and testes of both humans and experimental animals (Hee et al., 1988; Wang et al., 2017; Zhang et al., 2017; Zhong et al., 2022). While recent studies have focused on 1,2-DCE-induced central nervous damage, limited research has been conducted on its reproductive effects in male testes (Daniel et al., 1994; Zhang et al., 2017). In a previous study, we demonstrated that 28-day exposure to 1,2-DCE inhibits the cyclic adenosine monophosphate (cAMP)-response element binding protein (CREB)/cAMP-response element modulator (CREM) signaling cascade, resulting in testicular apoptosis, sperm malformation, and reduced sperm concentration in mice (Zhang et al., 2017). However, the underlying mechanisms by which 1,2-DCE disrupts the molecules responsible for reproductive toxicity in male testes remain largely unknown. Therefore, exploring the potential mechanisms of 1,2-DCE-induced male reproductive toxicity may offer valuable insights for identifying effective preventive measures.

A substantial portion of male reproductive injury induced by exogenous chemicals is characterized by testicular tissue cell death, including apoptosis, necrosis, and ferroptosis (Clair et al., 2012; Huang et al., 2009; Zhao et al., 2020). Pyroptosis is a programmed lytic cell death characterized by rapid plasma membrane disruption, DNA damage, and the release of proinflammatory intracellular content (Shi et al., 2015; Yan et al., 2021). Recent studies have demonstrated that exogenous chemical-induced damage to male testicular tissues exhibits features of pyroptosis in both mouse testes and GC-2 spd cells (Hong et al., 2021; Zhou et al., 2022). However, it remains to be explored whether 1,2-DCE induces male reproductive damage by triggering pyroptosis in testicular tissues.

P-element-induced wimpy (PIWI)-interacting RNAs (piRNAs) represent a class of germline-specific small noncoding RNAs isolated from mammalian germ cells (Girard et al., 2006; Grivna et al., 2006; Lau et al., 2006). PIWI-piRNA complexes are specifically expressed in animal germ cells and play essential roles in epigenetic regulation and germline development (Le Thomas et al., 2013; Wang et al., 2023). Both *in vitro* and *in vivo* studies have highlighted the significance of piRNA dysregulation in male testicular reproductive damage (Kong et al., 2021). Abnormal piRNA overexpression in mouse testes can lead to cell cycle arrest, abnormal sperm count, and morphology (Zhang et al., 2018a; Zhang et al., 2018c). However, the potential impact of 1,2-DCE on intracellular piRNA levels in testicular cells and the roles of altered piRNAs in 1,2-DCE-induced male testicular reproductive injury remain to be elucidated.

Melatonin, a hormone synthesized and secreted by the pineal gland, has also been found to be produced in the peripheral reproductive tract (Gonzalez-Arto et al., 2016). Evidence suggests that melatonin can cross the blood-testis barrier and be present in the testes (Zhang et al., 2018b; Zhang et al., 2018c). Several studies indicate that melatonin protects the

testes from chemotherapy regimens-induced damage, increases testosterone levels and improves the testicular microenvironment (Cao et al., 2021; Miao et al., 2022). Interestingly, recent findings have highlighted that the effects of melatonin concentrated on epigenetic modification (Lv et al., 2021; Miao et al., 2022). Building upon this knowledge, we hypothesized that melatonin could counteract the reproductive toxicity induced by 1,2-DCE through piRNA modifications.

To address the knowledge gap, we assessed the cell death modes induced by 1,2-DCE using death inhibitors targeting various pathways. We further verified the findings through quantitative polymerase chain reactions (qPCR), western blot, scanning electron microscopy (SEM), lactate dehydrogenase (LDH) assay, and flow cytometry. Additionally, we identified key piRNA and mRNA involved in 1,2-DCE-induced reproductive toxicity by conducting piRNA sequencing and transcriptome analysis in the mouse testes and GC-2 spd cells exposed to 1,2-DCE. Finally, we investigated the protective effect of melatonin on 1,2-DCE-induced testicular pyroptosis and its potential underlying mechanisms. Through these comprehensive analyses, we aimed to bridge the existing knowledge gaps regarding the protective effects of melatonin against 1,2-DCE-induced testicular pyroptosis, and highlight the potential therapeutic value of melatonin as a protective agent.

2. Materials and methods

2.1. Chemicals and reagents

We purchased 1,2-DCE (analytically pure, >99 % purity) from Guangzhou Chemical Reagent Factory (Guangzhou, China), and confirmed its structure with proton nuclear magnetic resonance. Melatonin (99.93 % purity), Caspase-1 (CASP-1) inhibitor VX-765, Gasdermin-D (GSDMD) inhibitor disulfiram, Bax inhibitor peptide V5, and autophagy inhibitor chloroquine were purchased from MedChemExpress (MCE, Shanghai, China) and diluted in dimethyl sulfoxide (DMSO) to prepare the stock solution (10 mM). Necroptosis inhibitor necrostatin-1 and necrosulfonamide (MCE) were diluted to a 20 mM concentration using DMSO, while PI3K inhibitor 3-methyladenine (MCE) was diluted to a 10 mM concentration using ultrapure water. Pan-caspase inhibitor Z-VAD-FMK, ferroptosis inhibitor ferrostatin-1, and deferoxamine mesylate (Selleck Chemicals, Shanghai, China) were diluted to a 10 mM concentration using DMSO. All other chemicals and solvents used were of molecular biology grade.

2.2. Animal and 1,2-DCE dynamic inhalation exposure

The Southern Medical University Scientific Research Committee on Ethics in the Care and Use of Laboratory Animals approved the animal experiments (SMUL2019037). Based on our preliminary studies, male mice were found to be more susceptible to 1,2-DCE exposure (Wang et al., 2017; Zeng et al., 2018). We obtained a total of 80 National Institutes of Health (NIH) Swiss male mice (specific pathogen free, 7 weeks old, body weight 18–20 g) from the Beijing Vital River Laboratory Animal Technology Co, Ltd (Beijing, China). Among them, 40 mice were used for 1,2-DCE exposure experiment, 40 mice were used for melatonin treatment study. All animals were housed and acclimated to the new environment in a climate-controlled (22 ± 2 °C) and humidity-specific (30–70 %) pathogen-free animal facility for 1 week. They had *ad libitum* access to food and water, and were kept under a 12-hour light/dark cycle (lights on at 8:00 a.m. and off at 8:00 p.m.).

For the 1,2-DCE exposure, we randomly assigned the animals into 4 groups, each containing 10 mice. They then underwent whole-body exposure to either dynamically inhaled 1,2-DCE or clean air (control group) using the HOPE-MED 8050F Inhalation Chamber System (Tianjin, China) for 6 h per day, consecutively for 28 days. Based on our preliminary studies, we designated the 1,2-DCE inhalational exposure levels as 0, 100, 350, and 700 mg/m³ for the male mice (Huang et al., 2020; Wang et al., 2017; Zeng et al., 2018; Zhang et al., 2017; Zhong

et al., 2020). During the 28-day consecutive exposure period, we conducted ongoing monitoring of 1,2-DCE concentrations within the exposure chambers. For the kinetic experiments, 60 mice were used for exposure quality control. This monitored the internal 1,2-DCE concentrations in the blood and testes during the exposure periods. Next, mice were exposed to 1,2-DCE or filtered room air for 6 h. The mouse blood and testes in each exposure group (0, 100, 350, 700 mg/m³) was collected at 3 time points (1,2-DCE exposure after 1 h and 6 h, and 1 h after stopping exposure), respectively. Air samples were collected from each chamber, and blood and testis samples were obtained from each individual mouse. We then conducted gas chromatography-mass spectrometry (GC-MS) analyses in accordance with the established protocols used in our previous studies (Zhang et al., 2017; Zhong et al., 2022). We used the time-weighted average (TWA) of 1,2-DCE concentration for the representative value of daily exposure level, as per equation: $TWA = [C_1T_1 + C_2T_2 + C_3T_3] \div [1 \text{ h/time} \times 3 \text{ times}]$, where C_1 is the 1,2-DCE concentration for time T_1 , C_2 is the 1,2-DCE concentration for time T_2 , etc. The real 1,2-DCE levels in the exposure chambers (0.16 ± 0.121 , 97.893 ± 31.243 , 357.021 ± 44.400 , and 689.071 ± 86.213 mg/m³) during the experiment closely matched the pre-established concentrations (0, 100, 350, and 700 mg/m³) (Table S1). In the mice, the internal 1,2-DCE exposure levels consistently increased in the 100, 350, and 700 mg/m³ groups, with a rapid ascent within the first hour, reaching their peak at 6 h of exposure. Subsequently, these levels decreased within an hour following the cessation of 1,2-DCE exposure (Table S1).

Within 24 h of their last exposure, the mice were anesthetized with a 3 % pentobarbital injection and euthanized. Testis and epididymis samples were collected from 10 mice in each group on ice for RNA and protein extraction. The testes were carefully dissected *en bloc* and weighed. Subsequently, the testes were frozen in liquid nitrogen and stored at -80°C . For gene expression microarray studies, we utilized the right side of the testis from each mouse. The left side of the testis from each mouse was divided into halves, with one half used for qPCR and the other half for western blot analysis.

2.3. Cell culture and 1,2-DCE treatment

We procured the mouse spermatogenesis pathway cell line GC-2 spd (ts) from American Type Culture Collection (CRL-2196, LGC Standards, UK). This cell line originated from freshly isolated spermatocytes (Hofmann et al., 1994), which had lost their differentiation potential and had been arrested at a premeiotic stage (Wolkowicz et al., 1996). GC-2 spd cells have been used extensively to study gene expression in germ cells, oxidative stress, pyroptosis and apoptosis (Choi et al., 2007; Hong et al., 2021; Solek et al., 2017; Solek et al., 2018). We grew this cell line in a high-glucose Dulbecco's modified Eagle's medium (DMEM, Gibco, Gaithersburg, MD, USA) supplemented with 10 % (v/v) fetal bovine serum (FBS, Gibco) and 1 % penicillin-streptomycin (Meilunbio, Dalian, China). For the mixed systems of 1,2-DCE exposures, we used 1 % DMSO and DMEM containing 10 % FBS and 1 % penicillin-streptomycin. The control group (0 mM) contained the same amount of DMSO. We maintained the GC-2 spd cells at 37°C in a humidified atmosphere with a 5 % CO₂-95 % air mixture. To ensure the absence of Mycoplasma contamination, we screened the GC-2 spd cell line using a One-step Quickcolor Mycoplasma Detection Kit (Yise Medical Technology, Shanghai, China) every 3 months. For experimental consistency, we performed all assays with GC-2 spd cells within the 10–20 passage range.

To determine the optimal concentration of 1,2-DCE for the *in vitro* study, we conducted a cell counting kit-8 (CCK-8) assay to assess the viability of GC-2 spd cells exposed to various concentrations of 1,2-DCE. We seeded GC-2 spd cells at a density of 8,000 cells/well in 96-well plates with plating medium. Subsequently, we treated the cells with 1,2-DCE at varying concentrations (0, 5, 10, 20, 30, 40, 50, 60, 70, 80 mM) in DMEM medium for 24 h. The control cells were treated with the same amount of 1 % DMSO alone. After incubation in a 5 % CO₂ incubator, we assessed the absorbance at 450 nm using the CCK-8 assay

(DOJINDO, Tokyo, Japan).

For subsequent experiments, we seeded GC-2 spd cells in a 6-well plate at a density of 4×10^5 cells/well and treated them with different concentrations of 1,2-DCE (0, 7.5, 15, 30 mM) for 24 h. The control group (0 mM) consisted of cells treated with DMEM containing 1 % DMSO, 10 % FBS, and 1 % penicillin-streptomycin. To investigate the mode of action of 1,2-DCE-induced cell death in GC-2 spd cells, we pre-treated cells exposed to 1,2-DCE (30 mM) with VX-765, disulfiram, Z-VAD-FMK, peptide V5, 3-methyladenine, chloroquine, necrosulfonamide, necrostatin-1, ferrostatin-1, or deferoxamine for 2 h. We then assessed cell viability using the CCK-8 assay.

To monitor the volatility of the 1,2-DCE in the media during the *in vitro* treatments, we collected the culture media from 7 time points (C_{0h}, C_{1h}, C_{2h}, C_{4h}, C_{8h}, C_{12h}, C_{24h}) after incubating the GC-2 spd cells with different concentrations of 1,2-DCE (0, 7.5, 15, 30 mM). We determined the 1,2-DCE levels in the culture media using GC-MS (Agilent) method following the Chinese National Standard GBZ/T 286-2016 (Zhong et al., 2022). Our results showed that 1,2-DCE concentrations had been declining in the culture media during the experiment (Table S2).

2.4. Melatonin treatment

Melatonin has been investigated for its potential as a therapeutic intervention against pyroptosis in reproductive toxicity (Luo et al., 2023). (1) In the *in vitro* study, we pretreated GC-2 spd cells with 125 μM melatonin for 2 h prior to subjecting them to 1,2-DCE (30 mM) treatment for 24 h in 6-well plates. At the end of the 24 h-treatments, we collected GC-2 spd cells for LDH release assays and western blot to evaluate the outcomes of melatonin treatment. (2) In the *in vivo* study, we randomly divided 40 male mice into four groups, each consisting of 10 mice: control, melatonin, 1,2-DCE and 1,2-DCE + melatonin. Melatonin was dissolved in normal saline at a concentration of 1 mg/mL and stored in a light-shielded environment at 4°C . The control and melatonin groups were to clean air for 6 h per day for 28 consecutive days, while the 1,2-DCE (700 mg/m³) and 1,2-DCE (700 mg/m³) + melatonin groups were subjected to dynamic inhalation of 1,2-DCE for the same duration. The 28-day dynamic inhalation exposure procedure was carried out as outlined in the previous section 2.2. In the 1,2-DCE + melatonin and melatonin groups, mice received a single oral dose of 20 mg/kg melatonin one hour before each dynamic inhalation session via gavage, while the control group and 1,2-DCE group received an oral of normal saline via gavage. The melatonin dose of 20 mg/kg was administered via gavage and was determined based on a comprehensive review of existing literature, considering previous studies that demonstrated effective results with this dosage in similar experimental setups (Akarca-Dizakar et al., 2020). This dose was selected to achieve a balance between efficacy and minimizing the potential for adverse effects. All groups were administered between 8:00 am and 8:30 am each day. At the end of the experiment, we collected testes and epididymides from all mice following deep anesthesia, perfusion, and euthanasia procedures.

We anesthetized each group of five mice with a 3 % pentobarbital injection and performed transcardial perfusion with ice-cold phosphate-buffered solution (PBS) until the right atrium was colorless. Subsequently, we transcardially perfused the mice in each group with 4 % paraformaldehyde (PFA). Following the collection of testis and caput epididymis tissue samples, we fixed them in 4 % (wt/vol) PFA at 4°C . The animals' testes were post-fixed in 4 % PFA overnight at 4°C , and subsequently embedded in paraffin wax. We embedded 5 mouse testes in each group in paraffin. Sections of 4 μm were collected, stained with hematoxylin-eosin (H&E), and mounted with neutral balsam. Photomicroscopic assessment of the testis tissue was conducted using a light microscope (Axiphot, Carl Zeiss, Oberkochen, Germany) at 400× magnification. Images were analyzed using Pannoramic MIDI (3D HISTECH, Budapest, Hungary) and Pannoramic Viewer (3D HISTECH) to detect any pathological lesions.

For sperm morphology assessment, we extracted sperm from the head epididymides of another set of anesthetized five mice in PBS solution. Subsequently, 10 μ L of sperm suspension was spread on slides, air-dried at room temperature, and fixed in methanol for 5 min. After air drying, the slides were examined using a light microscope. Employing a blinded methodology, three evaluators independently assessed 1000 spermatozoa from a single mouse, recording both abnormal morphological features and sperm quantity.

2.5. piRNA microarray analysis

To characterize the 1,2-DCE-associated piRNA changes in the mice, we built a compendium of genes related to the 1,2-DCE exposure using gene expression profiles from a piRNA sequencing study performed by OE Biotech Co. Ltd. (Shanghai, China). We extracted total RNA samples from the right testis tissues of male mice in the 0, 100, and 700 mg/m³ 1,2-DCE groups ($n = 3$ for each group) with Trizol reagent (Invitrogen, Carlsbad, CA, USA). We assessed RNA quantity and integrity using NanoDrop ND-2000 (Thermo Fisher Scientific, Waltham, MA, USA) and Agilent 2100 Bioanalyzer (Agilent technologies, Santa Clara, CA, USA). After RNA inspection, we followed the standard chip procedure for sample labeling, chip hybridization, and elution. Initially, we dephosphorylated and denatured total RNA and then labeled it with Cyanine-3-CTP (Cy3). Subsequently, we purified and hybridized the labeled RNA with the chip. After elution, we scanned the original image using the Agilent Scanner G2505C. To process the original image and extract the data, we utilized Feature Extraction (version 10.7.1.1, Agilent). Next, we conducted quantile standardization and subsequent processing with Genespring (version 13.1, Agilent). For data filtering, we employed probe markers and excluded reads that did not match with miRbase (<https://www.mirbase.org/>). Furthermore, we extracted the piRNA indices by searching piRBase (<https://www.regulatoryrna.org/database/piRNA/>). Using R (version 3.6.3) and the “Complexheatmap” package, we selected piRNAs as differentially expressed based on a P -value < 0.05 , $|\log_2 \text{fold change (FC)}| > \log_2 1.5$. The average basemean was in the top 10 %.

2.6. RNA transfection

Tsingke (Beijing, China) designed and synthesized the piRNA mimics and inhibitor to enhance and suppress the piRNA's function. To construct a plasmid overexpressing interferon regulatory factor 7 (*Irf7*), we digested the full-length mouse *Irf7* gene cDNA clone with NotI and XhoI (Thermo Fisher Scientific), and then cloned it into pCDNA3.1(+) to generate an *Irf7* overexpression plasmid. We purchased two independent small interfering RNAs (siRNAs) targeting the *Irf7* gene from Tsingke, and verified all of the constructions by sequence analysis. To initiate transfection, we seeded the GC-2 spd cells onto 6-well plates at approximately 50–70 % confluence, 24 h prior to transfection. Then, we utilized RNAFit (HanBio, Shanghai, China) for RNA transfection and diluted the piR-mmu-1019957 mimics and inhibitors to a concentration of 100 nM. Plasmid transfection was performed using 1.6 μ g/mL LipoFit 3.0 (HanBio). The sequences of the mimic, inhibitor and siRNA are listed in Table S3.

2.7. mRNA sequencing and gene differential expression analysis

To identify the target genes of piR-mmu-1019957, we treated GC-2 spd cells with 1,2-DCE (0 and 30 mM) and transfected them with piR-mmu-1019957 mimic and mimic control (100 nM). Then Shanghai Biotech Co. Ltd. (China) performed high throughput sequencing of the mRNA. We extracted total RNA following the manufacturer's instructions ($n = 3$ for each group). Total RNA was checked for an RIN number to inspect RNA integrity by an Agilent 2100 Bioanalyzer. Qualified total RNA was further purified by RNA Clean XP Kit (#A63987, Beckman Coulter, Inc, CA, USA) and RNase-Free DNase Set

(#79254, QIAGEN, GmBH, Germany). We assessed RNA quantity and integrity by NanoDrop ND-2000 (Thermo Fisher Scientific) and Agilent 2100 Bioanalyzer. To enrich and purify the mRNA, we depleted rRNAs from the total RNA. We used an Illumina NovaSeq6000 sequencer for sequencing, on the PE150 mode, generating a data volume of about 6G per sample. Then, we used an Illumina NovaSeq6000 sequencer on sequencing mode PE150. We normalized the expression levels of different samples using the trimmed mean of M values method, and converted the samples' normalized expression levels into fragments per kilobase of transcript per million mapped fragments (FPKM) (Robinson and Oshlack, 2010). We normalized the mRNA levels with significant modulation done by division, and then entered them into R (version 3.6.3) to draw a heatmap with the “Complex heatmap” package. We conducted GSEA with the “clusterProfiler” (Yu et al., 2012) (v. 3.14.0) package of R with databases from the Kyoto Encyclopedia of Genes and Genomes (KEGG), with the significance pathway set at $FDR \leq 0.1$ and $P \leq 0.05$.

2.8. SEM

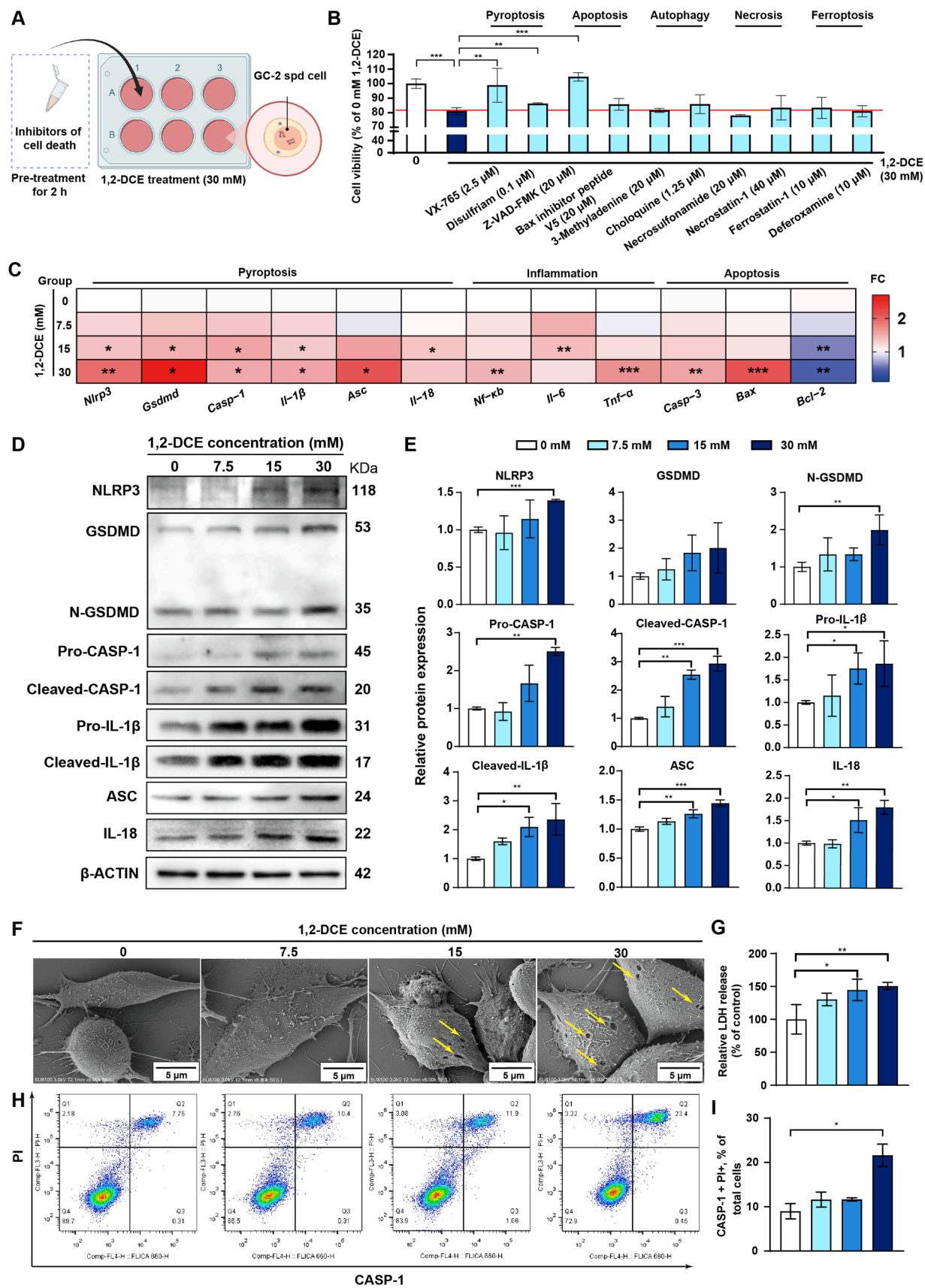
After treatment with varying concentrations of 1,2-DCE (0, 7.5, 15, 30 mM), we gently rinsed the GC-2 spd cells with PBS, then discarded the PBS and added electron microscope fixative solution (Servicebio, Wuhan, China) in the dark at room temperature for 2 h, followed by rinsing them with 0.1 % PBS three times. Then, we dehydrated them through a graded series of ethanol (30 %, 40 %, 50 %, 60 %, 70 %, 80 %, 90 %, and 100 %). After drying them in the CO₂ critical point dryer (K850, Quorum, East Sussex, UK), we adhered the samples to the sample stage with double-sided conductive tape and sprayed them with HITA-CHI MC1000 ion sputter (Hitachi, Tokyo, Japan). Then, we observed and photographed them by electron microscope. Finally, we sputter coated them with gold and examined them under a Hitachi Regulus 8100 scanning electron microscope operating at 15 kV.

2.9. qPCR

We homogenized the frozen mouse testes and cultured GC-2 spd cells in Trizol reagent (Thermo Fisher Scientific) using a high-speed low temperature homogenizer (Servicebio, Wuhan, China) at 60 Hz for 3 min at 4 °C. RNA quantity and purity were subsequently assessed using a NanoDrop ND-2000 (Thermo Fisher Scientific). Then, we synthesized the mRNA's cDNA with a 5 $\times$ Evo M-MLV RT Master Mix (Accurate Biotechnology, Hunan, China), and synthesized the piRNA's cDNAs with a first-strand cDNA synthesis super mix (Transgen, Beijing, China). Next, we performed qPCR detection for the mRNA, and piRNA levels for specific genes in the GC-2 spd cells using a 2 $\times$ SYBR®Green Realtime PCR Master Mix (Accurate Biotechnology) with a QuantStudio 6 Flex Real-Time PCR (Thermo Fisher Scientific). Tsingke designed and synthesized the primers (Table S4). We performed the reactions according to the following steps: 30 s at 95 °C, followed by 40 cycles of 5 s at 95 °C, and 30 s at 60 °C. We confirmed primer specificity by melting curve analysis, which showed a single product with the appropriate T_m for each primer set. Then, we analyzed all samples in quadruplicate and normalized the relative mRNA expression levels by β -actin, and the piRNA levels by *U6* expression in the same sample. Then, we calculated the relative mRNA and piRNA expressions using the 2^{− $\Delta\Delta C_t$} method (Livak and Schmittgen, 2001).

2.10. Western blot

We collected the GC-2 spd cells and washed them with pre-cooled PBS. Then we separately added the diced mouse testis tissues and GC-2 spd cells to ice-cold IP buffer (Aladdin, Los Angeles, CA, USA) containing 1:100 protease and phosphatase inhibitor cocktail (Keygen, Nanjing, China). After incubation for 30 min on ice, we centrifuged the homogenates at 13,000 $\times g$ for 30 min at 4 °C, and then collected the



(caption on next page)

Fig. 1. 1,2-DCE induced pyroptosis in the GC-2 spd cells in *in vitro* study. (A) Schematic diagram of the investigation of mode of action in 1,2-DCE-induced GC-2 spd cell death. (B) VX-765 (a Casp-1 inhibitor), disulfiram (a Gsdmd inhibitor), and Z-VAD-FMK (a pan-caspase inhibitor) rescued 1,2-DCE-induced GC-2 spd cell death ($n = 6$). (C) Key mRNA levels associated with pyroptosis, inflammation, and apoptosis in GC-2 spd cells altered by 1,2-DCE exposure ($n = 6$). (D, E) Upregulation of key proteins and their cleavage patterns related to pyroptosis in GC-2 spd cells in response to 1,2-DCE ($n = 3$). (F) SEM images illustrating microscopic changes in GC-2 spd cells treated with 1,2-DCE. The arrows indicate membrane pores in GC-2 spd cells in the 15 and 30 mM groups ($n = 3$). (G) *In vitro* exposure to 1,2-DCE increased LDH leakage from the culture medium of GC-2 spd cells ($n = 3$). (H, I) Flow cytometry identified caspase-1 and propidium iodide (PI) double-positive cells as indicative of pyroptosis. 1,2-DCE treatments caused a dose-dependent increase in the number and proportion of pyroptotic cells in GC-2 spd cells ($n = 3$). Comparisons were made with ANOVA, followed by a Tukey post hoc multiple comparisons test. We applied a Student's *t*-test to detect whether the differences between the two groups were statistically significant. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, vs. the control group or the indicated group.

supernatant. We quantified the supernatant's protein concentration with a Bradford Protein Assay Kit (Bio-Rad Laboratories, CA, USA). Next, we electrophoresed the proteins in each sample on a 12 % sodium dodecyl sulfate–polyacrylamide gel electrophoresis, and then transferred them onto a polyvinylidene fluoride membrane (Bio-Rad). Following blocking with 5 % (w/v) skim milk in Tris buffered saline tween for 2 h, we separately incubated the membranes with the primary antibodies β -ACTIN, IRF7, NOD-like receptor protein 3 (NLRP3), GSDMD, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-inflammatory factors interleukin-1 β (pro-IL-1 β), Cleaved-IL-1 β , inflammatory factors interleukin-18 (IL-18) and apoptosis-associated speck-like protein containing a caspase recruitment domain (ASC), overnight at 4 °C, and the secondary antibodies at room temperature for 2 h. Antibody information is listed in Table S5. We visualized protein bands with an enhanced chemiluminescence kit (Millipore, Billerica, MA, USA) and a Tanon-5200 chemical luminescence developing system (Tanon, Shanghai, China). Then, we performed gray analysis of the target protein bands with ImageJ 1.52v (NIH, Bethesda, MD, USA). We normalized all protein expression levels relative to each sample's β -ACTIN protein level, and then compared them to the control group.

2.11. LDH release assays

We determined cellular membrane damage by measuring the amount of LDH released in the cell culture supernatant. The LDH release was measured following the formation of formazon according to the LDH Assay kit (Nanjing Jiancheng Bioengineering Institute, Nanjing, China). In brief, we seeded GC-2 spd cells in 6-well plates. After the respective 1,2-DCE exposure, transfection treatment, or melatonin treatment, we collected the supernatant and performed LDH assays according to the manufacturer's instructions. We measured the absorbance at 450 nm using a Bio-Rad microplate reader.

2.12. Flow cytometry analysis

To assess pyroptosis in the GC-2 spd cells, we determined active caspase-1 using a FLICA 660 *in vitro* Casp-1 Detection kit (FLICA 660-YVAD-FMK, ImmunoChemistry Technologies, LCC, Bloomington, MN, USA). Additionally, we used propidium iodide (PI) stain, a fluorescent nucleic acid dye that only penetrates ruptured cell membranes, to label cells with pore formation. The cells (5×10^5 cells/mL) were resuspended in 295 μ L PBS and 5 μ L FLICA 660-YVAD-FMK dilution, and then incubated for 60 min in the dark. After the reaction, we washed the cells twice with 500 μ L wash buffer and discarded the supernatant. Then, 300 μ L wash buffer and 5 μ L PI were added to each tube to resuspend the cells, and gently mixed. The reaction was carried out for 5 min at 4 °C, while being protected from light. Finally, we analyzed the cells using a flow cytometer (AccuriC6, BD Biosciences, San Jose, CA, USA) with appropriate channels (FL3 channel for PI, FL4 channel for FLICA 660). Pyroptotic cells were defined as double positive for FLICA 660-YVAD-FMK and PI. The data were analyzed using FlowJo version 10.6.2 (FlowJo LLC, Ashland, OR, USA).

2.13. Statistical analysis

We presented the data as the mean $\pm$ standard deviation (SD). To

compare multiple groups, we conducted one-way analysis of variance (ANOVA) followed by a Tukey post hoc multiple comparisons test. For comparisons between two groups, we used a two-tailed Student's *t*-test, after confirming the homogeneity of variance. Data analysis was performed using SPSS 24.0 (IBM, Armonk, NY, USA), and statistical significance was considered at two-sided $P < 0.05$. Graphs were generated using GraphPad Prism 8.0 (San Diego, CA, USA).

3. Results

3.1. 1,2-DCE induced pyroptosis in GC-2 spd cells in *in vitro* study

In the pursuit of a comprehensive understanding of the precise mechanisms underlying 1,2-DCE-induced male reproductive damage, we conducted cytotoxicity assays using the mouse spermatocyte cell line, GC-2 spd. In the present study, the half maximal inhibitory concentration (IC₅₀) for 1,2-DCE in GC-2 spd cells is 46.36 mM. We found that 30 mM 1,2-DCE caused approximately 23 % death of GC-2 spd cells (Fig. S1A). To further elucidate the mode of action in 1,2-DCE-induced GC-2 spd cell death, we pretreated the cells with various cell death inhibitors 2 h before exposing them to 30 mM 1,2-DCE (Fig. 1A). We chose the maximum non-effective concentration of the various cell death inhibitors for subsequent rescue experiments in 1,2-DCE treatment (Fig. S1B–K). Our findings revealed that pyroptosis inhibitors, VX-765 and disulfiram, along with the pan-caspase inhibitor Z-VAD-FMK, effectively rescued the decrease in GC-2 spd cell activity caused by 1,2-DCE (Fig. 1B). Additionally, we observed that 1,2-DCE induced dose-dependent alterations in key mRNAs related to the signaling pathways of apoptosis, pyroptosis, and inflammation (Fig. 1C).

Next, we examined the expression of pyroptosis-related proteins to validate the observed changes in mRNA expressions. Notably, the protein levels of NLRP3, GSDMD, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-IL-1 β , Cleaved-IL-1 β , ASC, and IL-18 were all found to be upregulated in a dose-dependent manner with increasing concentrations of 1,2-DCE (Fig. 1D and E). Furthermore, SEM images revealed that 1,2-DCE dose-dependently formed membrane pores on the GC-2 spd cells (Fig. 1F). This observation was further supported by the increased LDH content in the cell culture medium supernatant, indicating enhanced GC-2 spd cell membrane breakage upon 1,2-DCE exposure (Fig. 1G). Flow cytometry results confirmed that 1,2-DCE treatment induced pyroptosis in GC-2 spd cells, as evidenced by the increased levels of caspase-1 (Fig. 1H and I). Collectively, these results provide compelling evidence that 1,2-DCE promotes testicular pyroptosis *in vitro* study.

3.2. 1,2-DCE-induced pyroptosis in testicular tissue and altered the piRNA profile in mice

To corroborate the observed phenomenon of 1,2-DCE-induced pyroptosis identified *in vitro*, we conducted a 28-day whole-body dynamic inhalation exposure of mice to 1,2-DCE (Fig. 2A). Subsequently, we validated the *in vitro* findings by demonstrating dose-dependent upregulations of pivotal pyroptosis-related proteins, e.g., NLRP3, GSDMD, N-GSDMD, Caspase-1, Cleaved-caspase-1, IL-1 β , Cleaved-IL-1 β , ASC, and IL-18, in the testes of male mice exposed to 1,2-DCE (Fig. 2B and C). These results provide compelling evidence that 1,2-DCE promotes testicular pyroptosis both *in vitro* and *in vivo*.

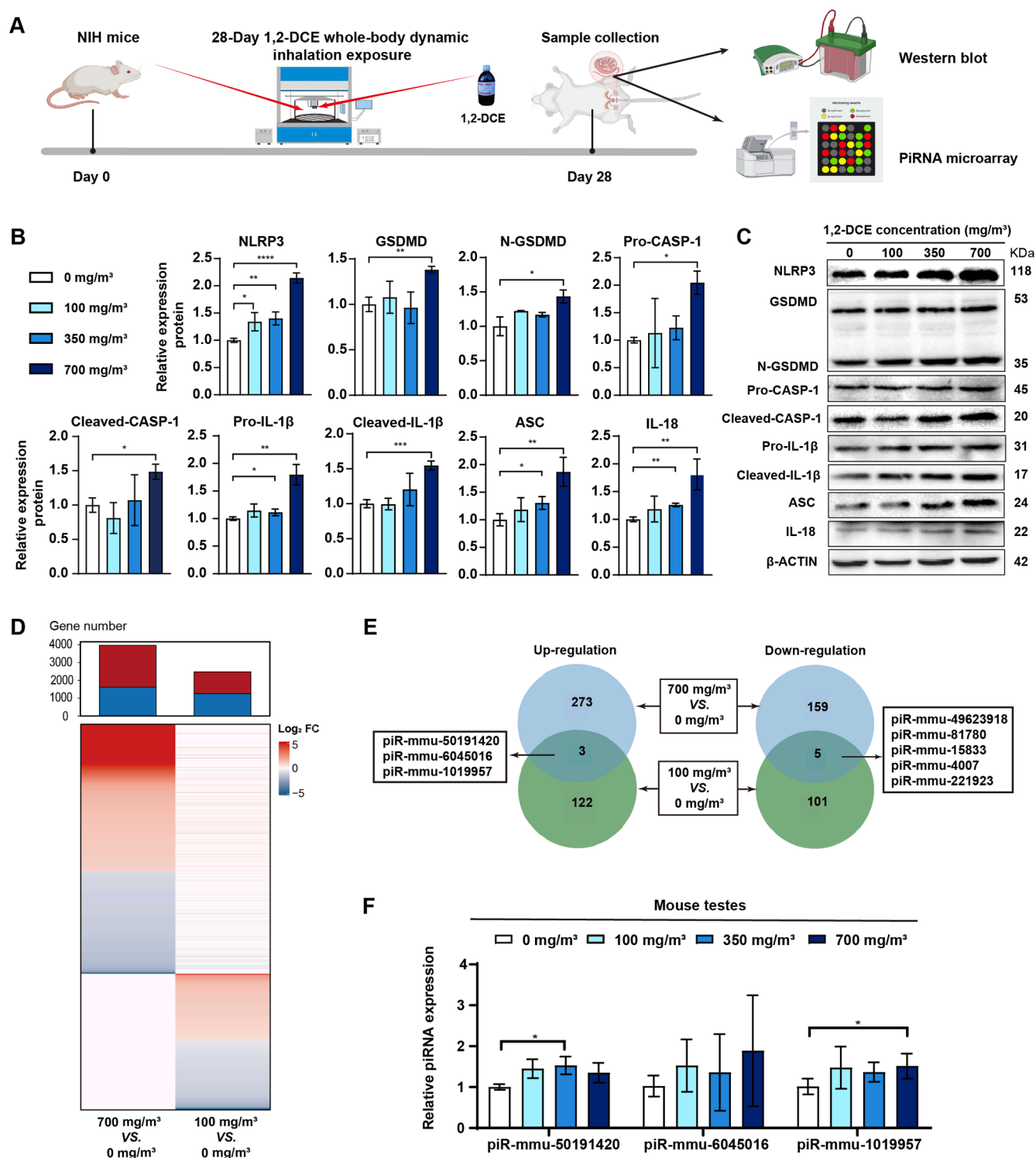
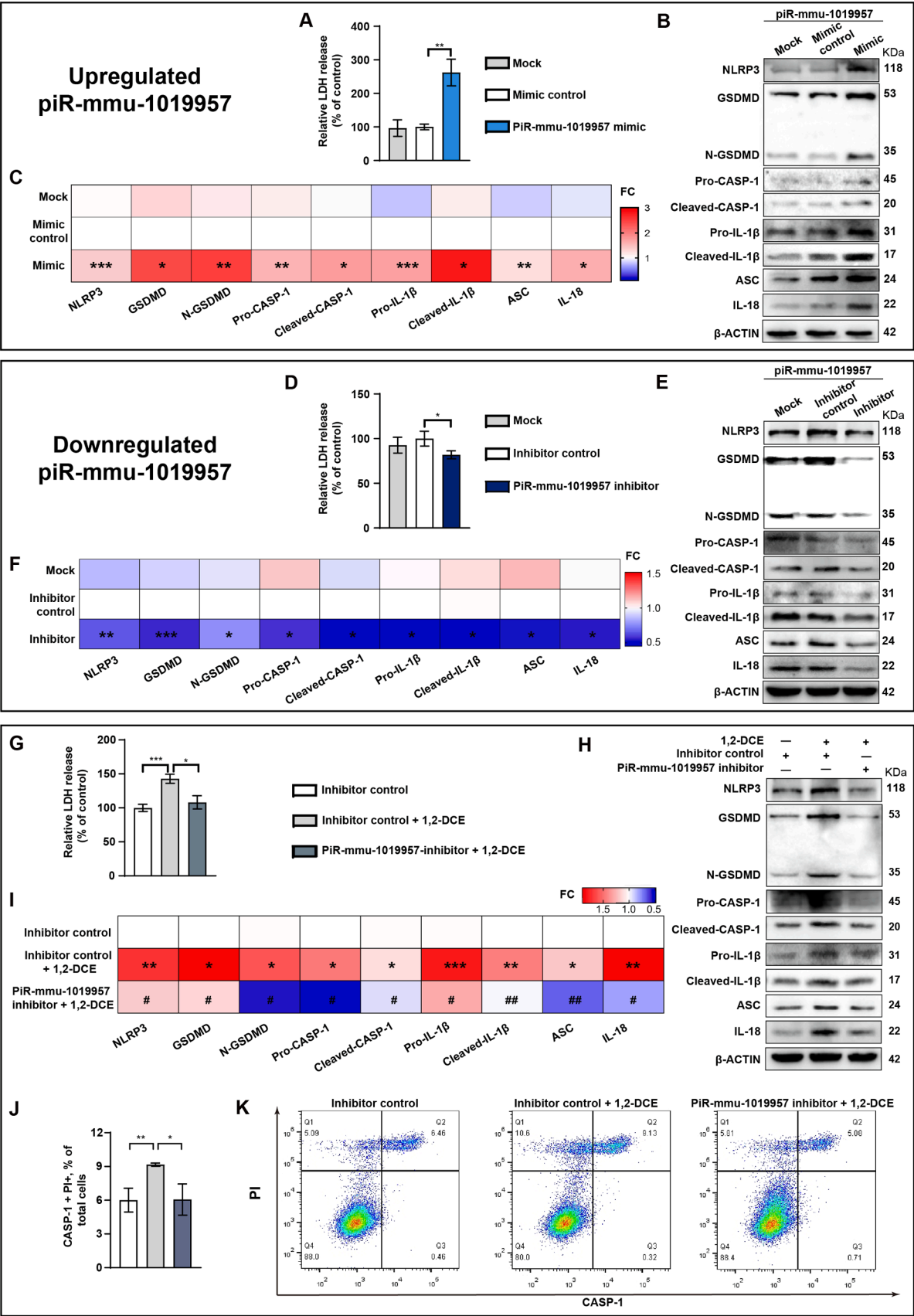


Fig. 2. 1,2-DCE induced pyroptosis in the mouse testes and altered the testicular piRNA profile in *in vivo* study. (A) Schematic diagram depicting the exploration of 1,2-DCE-induced pyroptosis in testicular tissue and the alteration of the piRNA profile in mice. (B, C) Increased levels of key pyroptosis-related proteins and their cleavage patterns in mouse testicular tissues after 1,2-DCE exposure ($n = 5$). (D) Heat map representing the altered piRNA expression profile in the mouse testes from the 100 mg/m³ and 700 mg/m³ 1,2-DCE exposure groups ($n = 3$). The color scheme indicated up-regulated genes (in red) and down-regulated genes. (E) PiRNA sequencing identified the common up- or downregulated piRNAs in the 100 and 700 mg/m³ 1,2-DCE groups ($n = 3$). (F) The three upregulated piRNAs from the sequencing results showed a consistent trend in the testicular tissues of mice ($n = 5$) exposed to 1,2-DCE. Comparisons were made with ANOVA, followed by a Tukey post hoc multiple comparisons test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, vs. the control group. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



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Fig. 3. PiR-mmu-1019957 induces pyroptosis in the GC-2 spd cells. (A) PiR-mmu-1019957 mimic increased LDH leakage from the culture medium of the GC-2 spd cells ($n = 3$). (B, C) PiR-mmu-1019957 mimic upregulated key pyroptosis-related proteins and their cleavage patterns ($n = 3$). (D) PiR-mmu-1019957 inhibitor decreased LDH leakage from the culture medium of the GC-2 spd cells ($n = 3$). (E, F) PiR-mmu-1019957 inhibitor downregulated key pyroptosis-related proteins and their cleavage patterns ($n = 3$). (G) PiR-mmu-1019957 inhibitor prevented increased LDH leakage from GC-2 spd cell cultures induced by 1,2-DCE treatment ($n = 3$). (H, I) PiR-mmu-1019957 inhibitor rescued the elevation of pyroptosis-associated proteins and their cleavage patterns induced by 1,2-DCE treatment ($n = 3$). (J, K) PiR-mmu-1019957 inhibitor mitigated the increase in the number and proportion of pyroptosis induced by 1,2-DCE treatment in the GC-2 spd cells ($n = 3$). We applied a Student's *t*-test to assess the statistical significance of the differences between the two groups. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, vs. the control group. # $P < 0.05$ and ## $P < 0.01$, vs. the Inhibitor control + 1,2-DCE group.

Furthermore, we analyzed piRNA expression profiles in mouse testes after 28 days of exposure to identify piRNAs involved in testicular pyroptosis following 1,2-DCE exposure. The results showed that 1258 piRNAs were upregulated and 1286 piRNAs were downregulated in the 100 mg/m³ 1,2-DCE group, while 2766 piRNAs were upregulated and 1647 piRNAs were downregulated in the 700 mg/m³ 1,2-DCE group, compared to the control. Heat maps were generated to illustrate the differentially expressed piRNAs in the mouse testes from the 100 mg/m³ and 700 mg/m³ exposure groups, in comparison to the control (Fig. 2D). Genes with a difference of ≥ 1.5 -fold in expression and basemean in the top 10 % were selected for subsequent investigation. As a result, we obtained 3 upregulated piRNAs (piR-mmu-50191420, piR-mmu-6045016, piR-mmu-1019957) and 5 downregulated piRNAs (piR-mmu-49623918, piR-mmu-81780, piR-mmu-15833, piR-mmu-4007, piR-mmu-221923), which exhibited altered expression levels in both the 100 mg/m³ and the 700 mg/m³ 1,2-DCE exposure groups (Fig. 2E, Table S6). We selected the 3 piRNAs (piR-mmu-50191420, piR-mmu-6045016, piR-mmu-1019957) with the highest fold change (FC) as the primary focus in the follow-up project. We found that all three piRNAs were upregulated in the testes of mice (Fig. 2F) and GC-2 cell lines (Figure S2) exposed to 1,2-DCE.

3.3. The role of piR-mmu-1019957 in 1,2-DCE-induced GC-2 spd cell pyroptosis in vitro study

To explore the functional association of these piRNAs with 1,2-DCE-induced testicular pyroptosis, we separately overexpressed them using piRNA mimics in the GC-2 spd cells (Fig. S3A–C). Among these, only the piR-mmu-1019957 mimic resulted in the upregulation of pyroptosis-related mRNAs, including *Nlrp3*, *Gsdmd*, *Caspase-1*, *Il-1 β* , *Asc* and *Il-18* (Fig. S3D). These findings indicated that 1,2-DCE induces alterations in testicular piRNA profiles, with piR-mmu-1019957 being associated with 1,2-DCE-induced testicular pyroptosis.

To elucidate the role of piR-mmu-1019957 in 1,2-DCE-induced pyroptosis, we investigated the expression of pyroptosis-related proteins in GC-2 spd cells transfected with piR-mmu-1019957 mimic for 24 h. The LDH content in the cell culture supernatant from the piR-mmu-1019957 overexpression group was higher than that in the control group (Fig. 3A). Additionally, the protein levels of NLRP3, GSDMD, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-IL-1 β , Cleaved-IL-1 β , ASC, and IL-18, were all found to be elevated after piR-mmu-1019957 overexpression (Fig. 3B and C). These results demonstrated increased pyroptosis observed in GC-2 spd cells upon piR-mmu-1019957 mimic treatment.

Next, to investigate the impact of piR-mmu-1019957 silencing in GC-2 spd cells, we transfected the cells with a piR-mmu-1019957 inhibitor for 24 h (Fig. S3E). The LDH content in the cell culture supernatant of the piR-mmu-1019957 inhibitor group was lower than that in the control group (Fig. 3D). In contrast to piR-mmu-1019957 overexpression, knockdown of piR-mmu-1019957 for 24 h resulted in significant downregulation of mRNA expression levels of *Nlrp3*, *Gsdmd*, *Caspase-1*, *Il-1 β* , *Asc* and *Il-18* (Fig. S3F). Similarly, the protein levels of NLRP3, GSDMD, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-IL-1 β , Cleaved-IL-1 β , ASC, and IL-18 were all reduced upon inhibition of piR-mmu-1019957 (Fig. 3E and F). These results demonstrated that the suppression of piR-mmu-1019957 attenuates pyroptosis in GC-2 spd cells.

We further substantiated the enhancing effect of piR-mmu-1019957

on 1,2-DCE-induced pyroptosis through rescue experiments. By employing the piR-mmu-1019957 inhibitor for 24 h, we successfully restored the altered levels of piR-mmu-1019957 induced by 1,2-DCE exposure for 24 h (Fig. S3G). Consequently, the piR-mmu-1019957 inhibitor successfully counteracted the 1,2-DCE-induced LDH leakage in GC-2 cells (Fig. 3G). Moreover, the piR-mmu-1019957 inhibitor effectively rescued the 1,2-DCE-induced upregulation of scorch-related mRNAs (e.g., *Nlrp3*, *Gsdmd*, *Caspase-1*, *Il-1 β* , *Asc* and *Il-18*) (Fig. S3H), and the corresponding proteins (e.g., NLRP3, GSDMD, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-IL-1 β , Cleaved-IL-1 β , ASC, and IL-18) (Fig. 3H and I). Flow cytometry analysis further confirmed the rescuing effect of the piR-mmu-1019957 inhibitor on 1,2-DCE-induced pyroptosis in GC-2 spd cells (Fig. 3J and K). Our findings collectively indicated that the piR-mmu-1019957 positively regulates IRF7, mediating 1,2-DCE-induced pyroptosis in GC-2 spd cells in *in vitro* study.

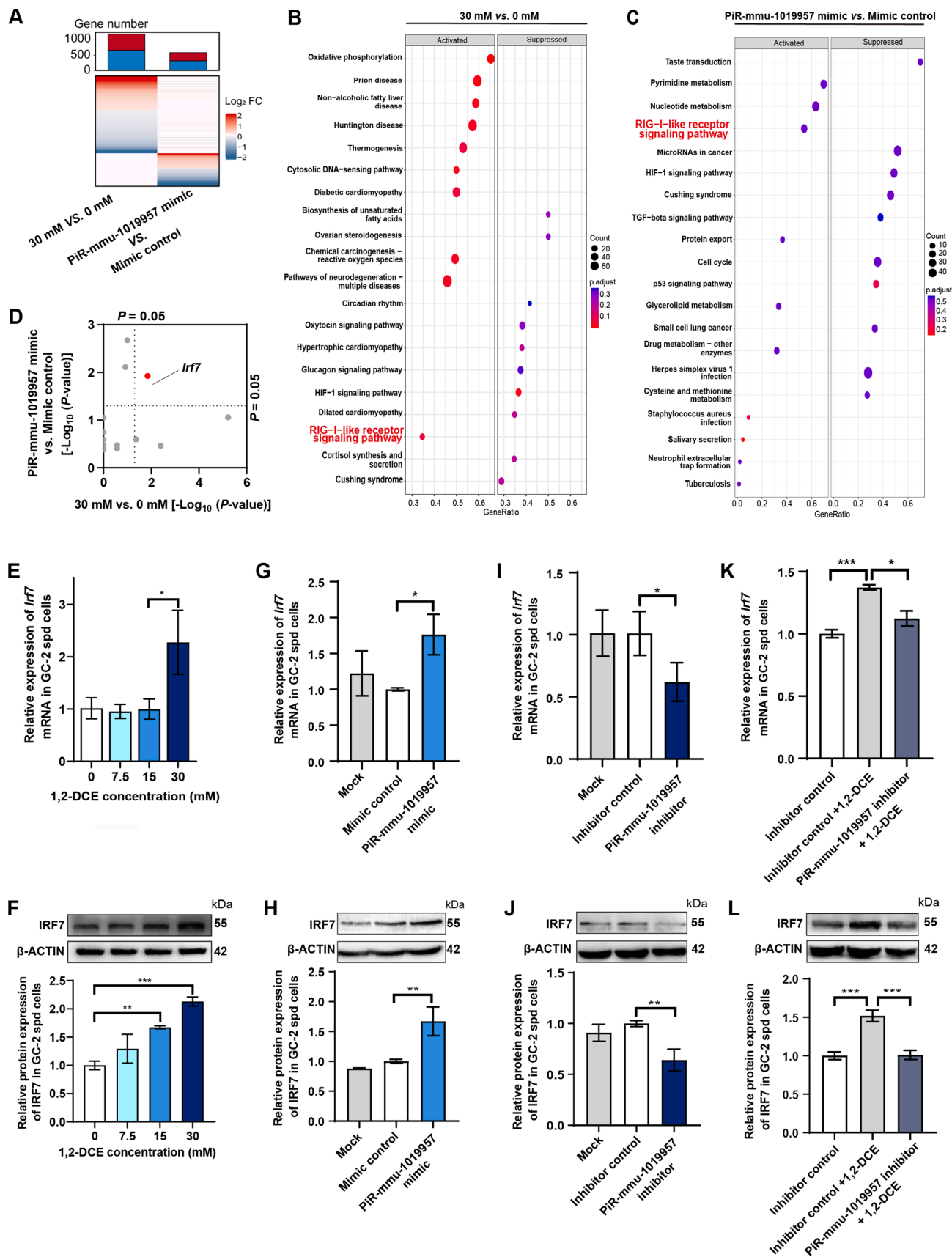
3.4. PiR-mmu-1019957 upregulated IRF7 levels in the 1,2-DCE-exposed GC-2 spd cells in vitro study

The downstream molecules regulated by piR-mmu-1019957 in 1,2-DCE-induced pyroptosis remained unidentified. To address this, we performed mRNA profiling of GC-2 spd cells with piR-mmu-1019957 overexpression and 1,2-DCE exposure (30 mM). The heat map illustrated the mRNA expression profile in the GC-2 spd cells subjected to 1,2-DCE exposure and piR-mmu-1019957 overexpression (Fig. 4A). Comparative analysis with the control group (0 mM) revealed 529 upregulated and 662 downregulated genes in the 30 mM 1,2-DCE group ($P < 0.05$) (Fig. 4A). In the piR-mmu-1019957 overexpression group, there were 269 upregulated genes and 315 downregulated genes compared to the mimic control ($P < 0.05$) (Fig. 4A).

Next, we performed GSEA pathway enrichment analysis by selecting genes with similar change pattern in both the 1,2-DCE exposure group and the piR-mmu-1019957 overexpression group (Fig. 4B and C). The bubble plots revealed that the RIG-I-like receptor signaling pathway was activated in both groups (Fig. 4B and C). Among the altered genes enriched in the RIG-I-like receptor signaling pathway, *Irf7* was found to be the most upregulated gene in both groups ($P < 0.05$) (Fig. 4D). We found an increase in IRF7 mRNA and protein in both the GC-2 spd cells and the mice exposed to 1,2-DCE (Fig. 4E and F and Figure S4). Moreover, piR-mmu-1019957 mimic led to the upregulation of IRF7 mRNA and protein (Fig. 4G and H). Conversely, piR-mmu-1019957 inhibitor caused downregulation of IRF7 mRNA and protein (Fig. 4I and J); PiR-mmu-1019957 inhibitor was able to rescue the 1,2-DCE-induced upregulation of IRF7 mRNA and protein levels (Fig. 4K and L). Taken together, these results provided evidence that IRF7 is a downstream target of piR-mmu-1019957.

3.5. IRF7's role in 1,2-DCE-induced GC-2 spd cell pyroptosis

To investigate the roles of IRF7 in 1,2-DCE-induced pyroptosis, we upregulated the IRF7 levels in the GC-2 spd cells by transfecting *Irf7* overexpression plasmids for 24 h (Fig. S5A). The results showed that increased LDH leakage in the GC-2 spd cells after IRF7 overexpression (Fig. 5A). Additionally, we found that the *Irf7* overexpression plasmid caused elevated levels of IRF7 and pyroptosis-related proteins, e.g., NLRP3, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-IL-1 β , Cleaved-IL-1 β , ASC and IL-18 (Fig. 5B and C). Then, we silenced *Irf7* in the GC-2 spd



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Fig. 4. PiR-mmu-1019957 upregulated IRF7 levels in the 1,2-DCE-exposed GC-2 cells. (A) Heat map depicting the altered mRNA expression profiles in the GC-2 spd cells with piR-mmu-1019957 overexpression and 1,2-DCE exposure (30 mM) ($n = 3$). The color scheme indicated up-regulated genes (in red) and down-regulated genes. (B) Bubble plot of the top 20 pathway enrichment analysis entries for mRNAs with consistent trends of upregulation and downregulation within the 1,2-DCE exposure and (C) piR-mmu-1019957 mimic groups. The RIG-I-like receptor signaling pathway was activated in both groups. (D) *Irf7*, enriched in the RIG-I-like pathway, was most strongly upregulated in both groups. (E, F) IRF7 mRNA and protein levels increased along with the increasing 1,2-DCE concentrations in the GC-2 spd cells ($n = 3$). (G, H) PiR-1019957 mimic upregulated IRF7 mRNA and protein. (I, J) PiR-1019957 inhibitor downregulated IRF7 mRNA and protein. (K, L) PiR-1019957 inhibitor rescued the upregulated IRF7 mRNA and protein caused by 1,2-DCE. Comparisons were made with ANOVA, followed by a Tukey post hoc multiple comparisons test. We applied a Student's *t*-test to detect whether the differences between the two groups were statistically significant. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, vs. the control group or the indicated group. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

cells by siRNA for 24 h (Fig. S5B and C). We found that LDH leakage decreased in the GC-2 spd cells after silencing *Irf7* (Fig. 5D). Consequently, the levels of IRF7 and pyroptosis-related proteins, such as NLRP3, GSDMD, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-IL-1 β , Cleaved-IL-1 β , ASC, and IL-18, had decreased in response to si-*Irf7* (Fig. 5E and F). Furthermore, si-*Irf7* inhibited 1,2-DCE-induced increase in LDH leakage in the GC-2 spd cells (Fig. 5G). Also, si-*Irf7* treatment inhibited the 1,2-DCE-induced elevation of IRF7 and pyroptosis-related proteins (e.g., NLRP3, GSDMD, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-IL-1 β , Cleaved-IL-1 β , ASC, and IL-18) in the GC-2 spd cells (Fig. 5H and I). Finally, flow cytometry showed that si-*Irf7* treatment had rescued 1,2-DCE-induced pyroptosis (Fig. 5J and K). These results demonstrated that 1,2-DCE induces pyroptosis by elevating IRF7 in the GC-2 spd cells.

3.6. PiR-mmu-1019957 promoted 1,2-DCE-induced GC-2 spd cell pyroptosis through the upregulation of IRF7

To further validate the regulatory effect of piR-mmu-1019957 on IRF7, we co-transfected piR-mmu-1019957 mimic and si-*Irf7* in the GC-2 spd cells (Fig. 6A). The results showed that si-*Irf7* mitigated the LDH leakage induced by piR-mmu-1019957 mimic in the GC-2 spd cells (Fig. 6B). Also, si-*Irf7* inhibited the upregulation of IRF7 and pyroptosis-related proteins (e.g., NLRP3, GSDMD, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-IL-1 β , Cleaved-IL-1 β , ASC, and IL-18) induced by piR-mmu-1019957 mimic (Fig. 6C and D). Finally, the flow cytometry results showed that si-*Irf7* rescued the GC-2 spd cell pyroptosis activated by piR-mmu-1019957 mimic (Fig. 6E and F). In sum, these findings suggest that IRF7 plays a critical role in the promotion of 1,2-DCE-induced GC-2 spd cell pyroptosis through the upregulation of piR-mmu-1019957.

Melatonin attenuates pyroptosis by inhibiting piR-mmu-1019957/IRF7 pathway in 1,2-DCE-induced reproductive toxicity.

To evaluate the potential protective effects of melatonin against 1,2-DCE-induced pyroptosis and reproductive injury, melatonin was administered prior to 1,2-DCE treatment *in vitro* (Fig. 7A). Our results demonstrated a significant reduction in the upregulation of piR-mmu-1019957 and the restoration of decreased activity of 1,2-DCE-induced GC-2 spd cells following pretreatment with a 125 μ M dose of melatonin (Fig. 7B, Figure S6). Therefore, we employed a 125 μ M melatonin treatment in the subsequent *in vitro* investigations. In accordance with the finding, melatonin effectively restored protein levels of IRF7 and pyroptosis-related markers, including NLRP3, GSDMD, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-IL-1 β , Cleaved-IL-1 β , ASC, and IL-18, in GC-2 spd cells exposed to 1,2-DCE (Fig. 7C and D). This was accompanied by a reduction in LDH release (Fig. 7E).

We subsequently conducted animal experiments to corroborate the rescuing effects of melatonin identified in the *in vitro* study (Fig. 8A). Consistent with the *in vitro* results, we observed that melatonin effectively mitigated 1,2-DCE-induced atrophy of the germinal epithelium in testicular sections of mice (Fig. 8B), which was accompanied by a reduction in sperm abnormalities (Fig. 8C). Representative images depicting sperm abnormalities in mice exposed to 1,2-DCE at 700 mg/ m^3 could be found in the Figure S7. Importantly, the protective effect is

primarily exerted by inhibiting the IRF7-mediated pyroptosis pathway (Fig. 8D and E). Collectively, these results indicated that melatonin confers protection against 1,2-DCE-induced pyroptosis and reproductive toxicity by inhibiting the piR-mmu-1019957/IRF7 pathway.

4. Discussion

Our previous study has demonstrated that 1,2-DCE induces reproductive toxicity in male NIH mice via the CREM/CREB pathway (Zhang et al., 2017). However, the precise mechanisms by which 1,2-DCE disrupts molecules causing reproductive toxicity in male testes remain unclear. Moreover, the potential regulatory role of piRNAs in this process remains to be elucidated. To address the knowledge gap, we established an *in vitro* model to investigate the phenotype of GC-2 cell death induced by 1,2-DCE exposure. Our study revealed that exposure to 1,2-DCE induced cell death *in vitro*, which can be rescued by VX-765 and disulfiram. The induction of pyroptosis was evidenced by dose-dependent increases in pyroptosis-related mRNA, protein levels, and their cleavage patterns. Furthermore, both pore formation on GC-2 spd cell membranes and LDH leakage exhibited dose-dependent increments. Moreover, the number and proportion of pyroptotic cells increased with rising 1,2-DCE concentrations. Intrigued by the potential regulatory role of piRNAs, we examined the expression profile of piRNAs in the testes of mice exposed to 1,2-DCE. We found that piR-mmu-1019957 was significantly overexpressed both *in vitro* and *in vivo*, and its upregulation positively regulated the pyroptosis of GC-2 spd cells. Further investigations led to the discovery that piR-mmu-1019957 was responsible for 1,2-DCE-induced pyroptosis in GC-2 spd cells through the upregulation of IRF7 in the RIG-I-like signaling pathway. Strikingly, we observed that melatonin modulated the piR-mmu-1019957-mediated regulation of IRF7, thereby mitigating pyroptosis in the context of 1,2-DCE-induced testicular toxicity. Notably, despite the previous evidence of 1,2-DCE-mediated reproductive toxicity via the CREM/CREB pathway, we did not observe co-regulation of *Creb* and *Crem* transcript levels by 1,2-DCE and piR-mmu-1019957 at the cellular level. This observation suggests that 1,2-DCE might primarily act through alternative pathways when directly influencing GC-2 spd cells *in vitro*. In this context, it is evident that piR-mmu-1019957 assumes a crucial role in promoting 1,2-DCE-induced testicular pyroptosis by upregulating IRF7 in GC-2 spd cells.

In recent studies, several exogenous chemicals have been found to impair mouse germ cells through the CASP-1/GSDMD classical pyroptosis pathway (Hong et al., 2021; Zhou et al., 2022). In our current investigation, we utilized specific pyroptosis inhibitors, VX-765 targeting CASP-1, and disulfiram targeting GSDMD, to demonstrate that 30 mM 1,2-DCE induces cell death in GC-2 spd cells *in vitro*, and this effect can be rescued by these inhibitors. Further experiments confirmed that 1,2-DCE exposure induces pyroptosis in GC-2 spd cells, a finding consistent with pyroptosis observed in the testicular tissues of mice exposed to 1,2-DCE via inhalation. The successful rescue of pyroptosis by VX-765 and disulfiram confirmed that 1,2-DCE indeed triggers pyroptosis in mouse testicular tissues and GC-2 spd cells. Additionally, we observed a decrease in cell activity induced by 30 mM 1,2-DCE, which could be partially saved by Z-VAD-FMK. Z-VAD-FMK is a broad-spectrum caspase inhibitor, capable of inhibiting pyroptosis-related

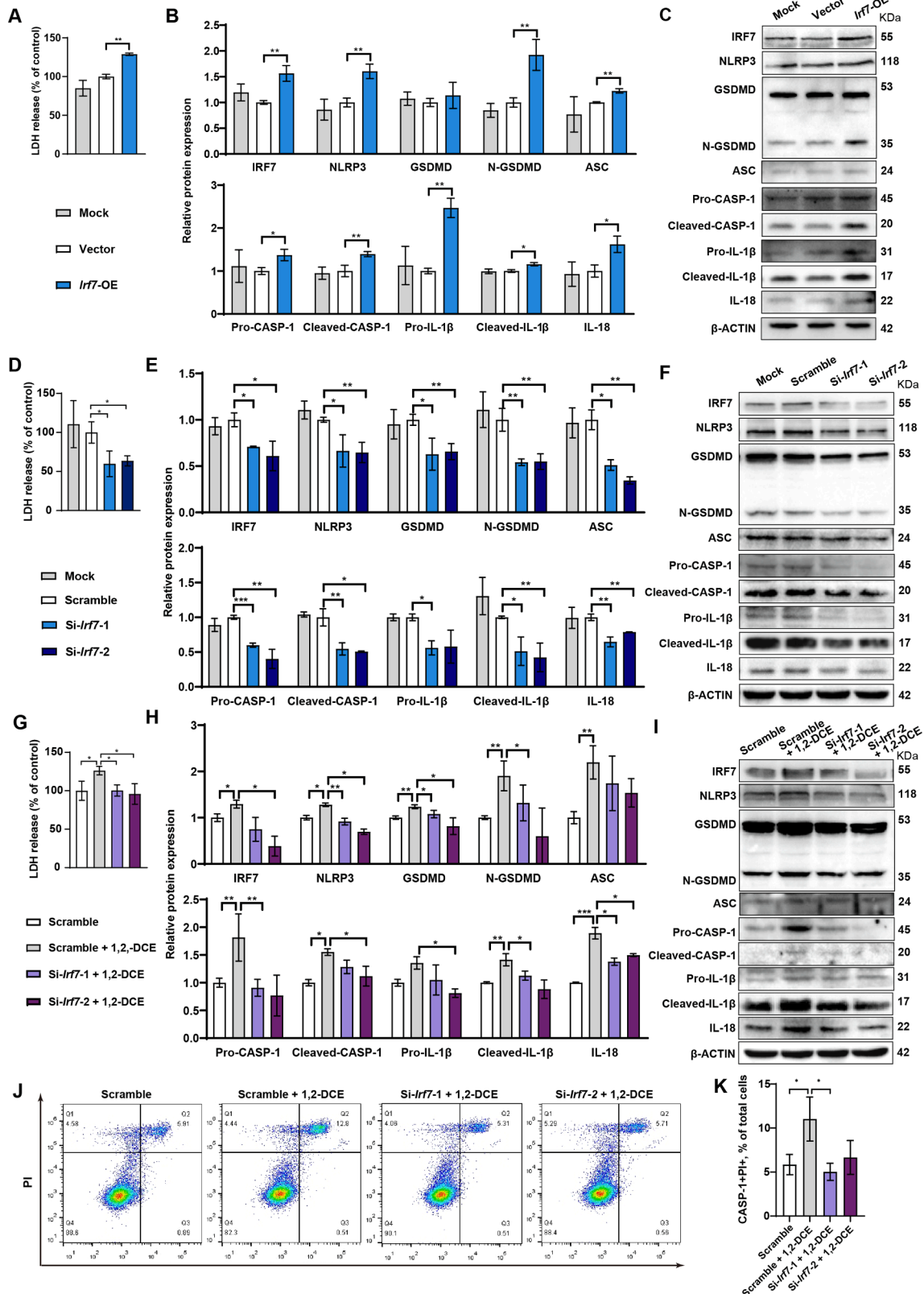


Fig. 5. IRF7 caused pyroptosis in the GC-2 spd cells. (A) LDH leakage from the culture medium of the GC-2 spd cells increased after IRF7 overexpression ($n = 3$). (B, C) The *Irf7* overexpression plasmid caused elevated levels of IRF7, pyroptosis-related proteins and their cleavage patterns ($n = 3$). (D) LDH leakage from the culture medium of the GC-2 spd cells decreased after si-*Irf7* transfection ($n = 3$). (E, F) Si-*Irf7* treatment mitigated the expression levels of IRF7, pyroptosis-related proteins and their cleavage patterns ($n = 3$). (G) Si-*Irf7* reversed the increased LDH leakage from the GC-2 spd cell cultures induced by 30 mM 1,2-DCE ($n = 3$). (H, I) Si-*Irf7* prevented the elevation of IRF7, pyroptosis-associated proteins and their cleavage patterns induced by 30 mM 1,2-DCE in the GC-2 spd cells ($n = 3$). (J, K) Si-*Irf7* saved the increase in the proportion of pyroptosis cells induced by 30 mM 1,2-DCE ($n = 3$). We applied a Student's *t*-test to assess the statistical significance of the differences between the two groups. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, vs. the indicated group.

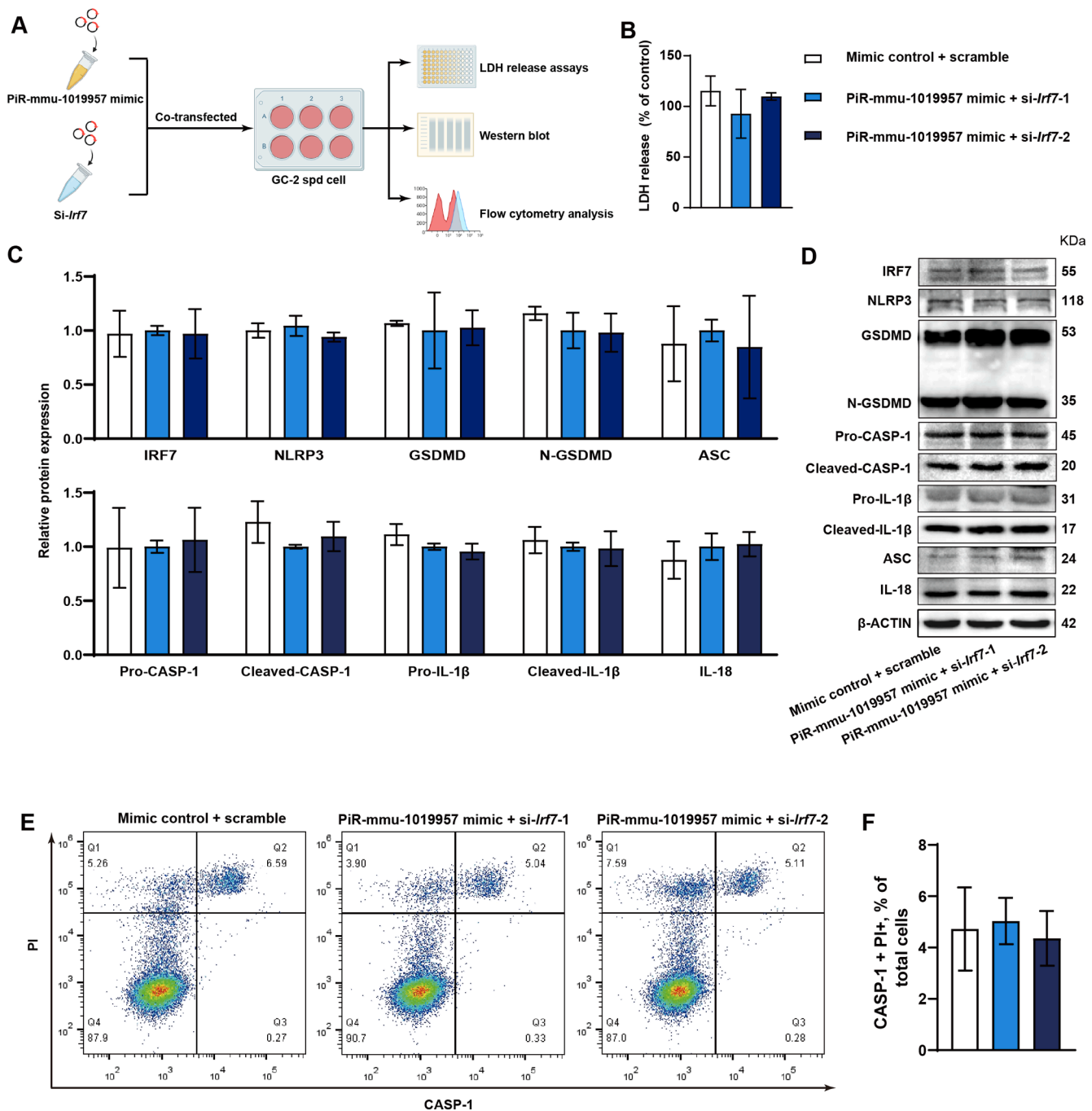


Fig. 6. piR-mmu-1019957 promoted 1,2-DCE-induced pyroptosis through IRF7 upregulation in the GC-2 spd cells. (A) Schematic diagram illustrating the experimental design to validate the regulatory effect of piR-mmu-1019957 on IRF7. (B) Si-Irf7 attenuated piR-mmu-1019957 mimic-induced LDH leakage in GC-2 spd cells ($n = 3$). (C, D) Si-Irf7 inhibited the upregulation of IRF7 and pyroptosis-related proteins induced by piR-mmu-1019957 mimic ($n = 3$). (E, F) Si-Irf7 rescued the piR-mmu-1019957 mimic-activated GC-2 spd cell pyroptosis ($n = 3$). We applied a Student's t -test to assess the statistical significance of the differences between the two groups.

CASP-1 and apoptosis-related CASP-3 and CASP-7 (Graczyk, 2002). In this study, exposure to 1,2-DCE *in vitro* induced changes in apoptosis-related mRNA, such as increased Casp-3 and Bax levels, and decreased Bcl-2 levels, indicating the induction of apoptosis in GC-2 spd cells. The decrease in cell activity induced by 30 mM 1,2-DCE could be partially rescued by the apoptosis inhibitor Bax inhibitor peptide V5, although the difference was not statistically significant. Bax-inhibiting peptide V5 is a Bax-mediated apoptosis inhibitor that can down-regulate BAX protein levels by 10 % (Rivas-Carrillo et al., 2007). Previous studies have shown that 1,2-DCE induces testicular tissue apoptosis in mice after

inhalation of dynamic toxicity *in vivo* (Zhang et al., 2017). However, the present study results revealed that 1,2-DCE primarily induced pyroptosis of GC-2 spd cells when exposed *in vitro*, followed by inflammation and apoptosis.

PiRNAs, a class of small non-coding RNAs, are highly expressed in germ cells, forming PIWI-piRNA complexes with PIWI protein family members. Study on piRNA's role in reproductive toxicity has primarily focused on epigenetic regulation. PIWIs/piRNAs can influence gene function both before and after transcription (Kiuchi et al., 2014; Pezic et al., 2014; Rajasethupathy et al., 2012). Pretranscriptional regulation

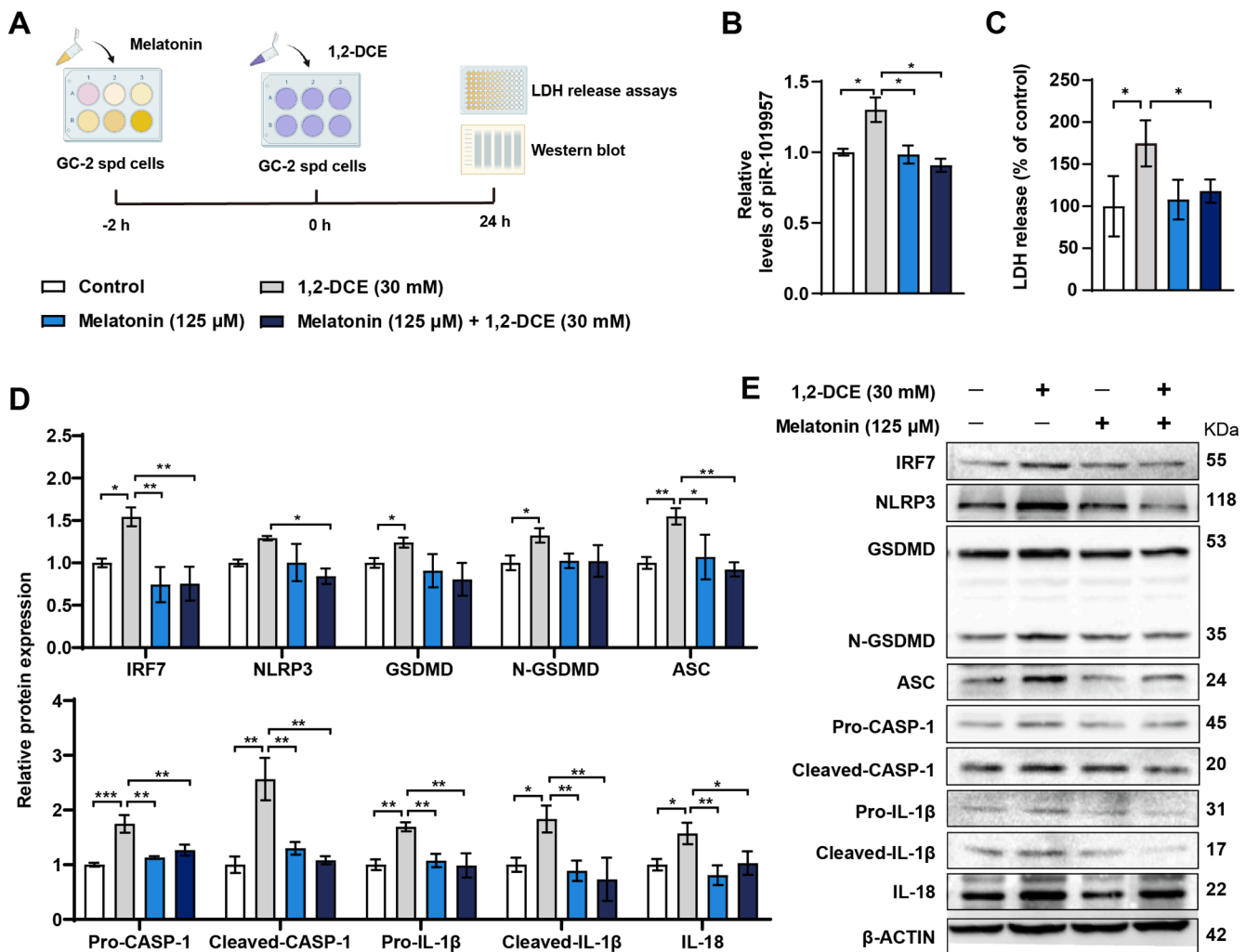


Fig. 7. Effects of melatonin supplementation on GC-2 spd cells exposed to *in vitro* 1,2-DCE-induced pyroptosis. (A) Schematic diagram illustrating the experimental design for assessing the potential protective effects of melatonin against 1,2-DCE-induced pyroptosis. (B) Melatonin pretreatment significantly reduced the upregulation of piR-mmu-1019957 in GC-2 spd cells exposed to 1,2-DCE ($n = 3$). (C) Melatonin effectively restored LDH release in GC-2 spd cells treated with 1,2-DCE ($n = 3$). (D, E) Melatonin effectively restored protein levels of IRF7 and pyroptosis-related markers, including NLRP3, GSDMD, N-GSDMD, pro-CASP-1, Cleaved-CASP-1, pro-IL-1β, Cleaved-IL-1β, ASC, and IL-18, in GC-2 spd cells exposed to 1,2-DCE ($n = 3$). Comparisons were made with ANOVA, followed by a Tukey post hoc multiple comparisons test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, vs. the control group or the indicated group.

involves inhibition of target production through chromatin modifications, such as histone or DNA methylation (Aravin et al., 2008; Sienski et al., 2012). Post-transcriptional regulation occurs when PIWI/piRNAs bind to the 3'-UTR of target mRNAs through base complementary pairing, leading to mRNA degradation and down-regulation, thereby affecting the function of downstream target genes. Recent studies have revealed that piRNAs can have positive regulatory roles on downstream mRNAs. For instance, piR-823 increases the expression of HSP27, HSP60, and HSP70 by enhancing the transcriptional activity of the transcription factor HSF1 (Yin et al., 2017). Inhibition of piRNA-30473 leads to a significant reduction in WTAP mRNA and protein levels, as piRNA-30473 stabilizes WTAP by targeting complementary sequences in the non-coding region (Han et al., 2021). Our study demonstrated that piR-mmu-1019957 overexpression increases the mRNA and protein levels of *Irf7*, while the piRNA inhibitor prevents the rise of *Irf7* levels induced by 1,2-DCE in GC-2 spd cells. These findings offer a novel perspective on piRNA function. Studies have suggested that piRNAs can upregulate mRNA post-transcriptional levels by mediating intronic cleavage of mRNA precursors. For instance, piRs-3-13 promotes precursor NPF1 splicing into a mature form, leading to a significant increase in NPF1 expression (Wang et al., 2022). However, whether piRNA upregulates *Irf7* mRNA transcription by promoting its maturation in our

study remains uncertain and requires further investigation.

IRF7, initially identified as a lymphoid-specific factor, is primarily expressed in fibroblasts, B cells, lymphocytes, pDCs, and monocytes. It plays a crucial role in regulating inflammatory responses, cell differentiation, and apoptosis (Jiang et al., 2014). Additionally, IRF7 directly binds to GSDMD and its overexpression promotes adipocyte pyroptosis, leading to increased CAPS-1 and IL-1β expression levels (Liu et al., 2017). As a key protein in the RIG-I-like cell signaling pathway, IRF7 has significant regulatory functions (Deng et al., 2022; Liu et al., 2022a, 2022b), which is closely related to pyroptosis. For instance, myoglobin induces macrophage pyroptosis through the RIG-I/CASP-1/GSDMD signaling pathway in CS-AKI (Li et al., 2022). Our study demonstrated that 1,2-DCE exposure and piR-mmu-1019957 mimic increased IRF7 mRNA and protein levels, subsequently upregulating pyroptosis-related proteins to induce pyroptosis in GC-2 spd cells. These findings support existing literature and underscore the crucial regulatory role of IRF7 in 1,2-DCE-induced pyroptosis in GC-2 spd cells. Nevertheless, whether piR-mmu-1019957 directly regulates *Irf7* mRNA levels requires further exploration, and it will be the focus of our future research.

The beneficial effects of melatonin on spermatogenic cells under pathological conditions have been consistently reported in previous studies, establishing its potential as a protective agent (Cao et al., 2021;

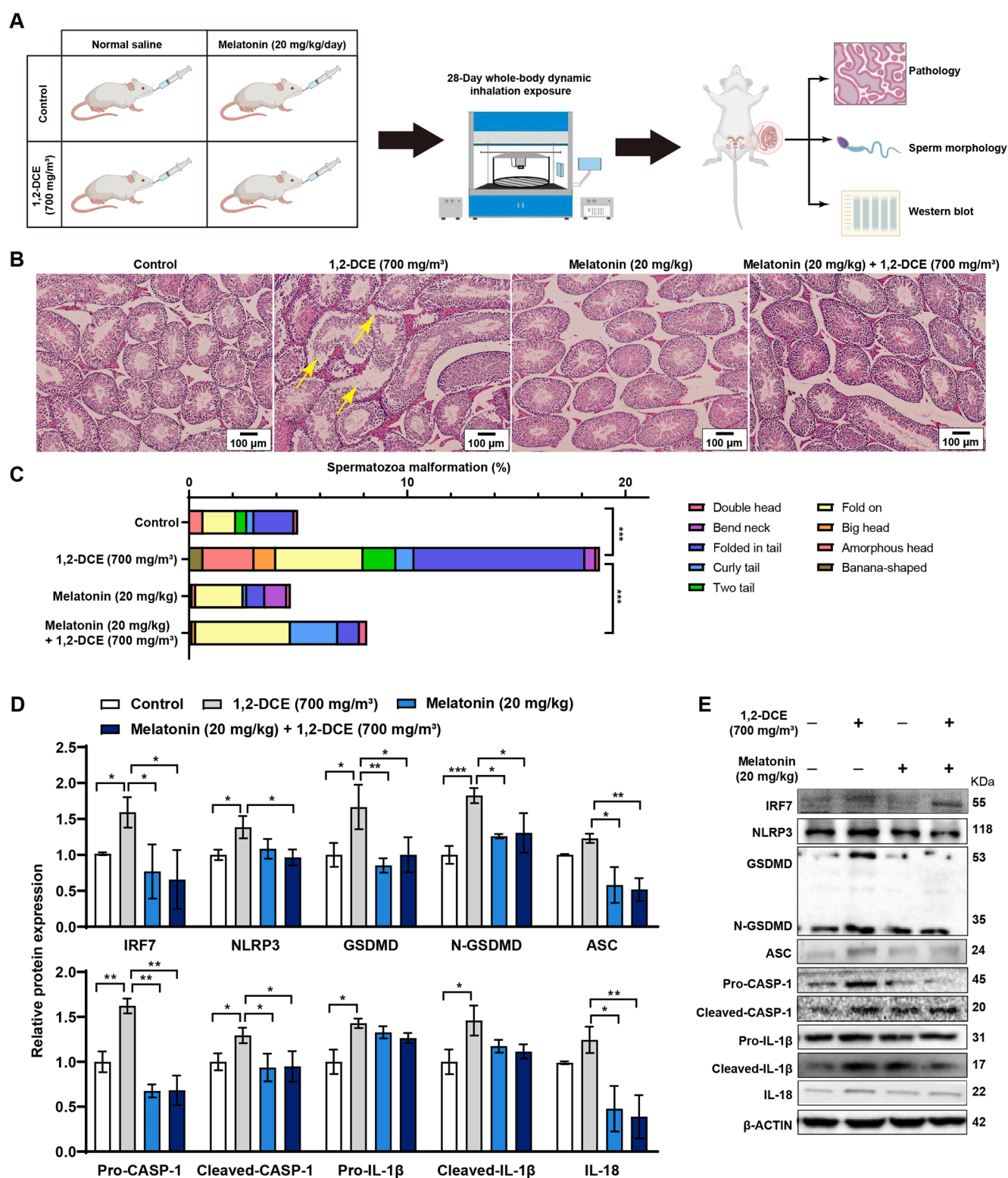


Fig. 8. Effects of melatonin supplementation on spermatogenesis pathway cells in mice exposed to 1,2-DCE-induced reproductive toxicity *in vivo*. (A) Schematic diagram illustrating the experimental design for assessing the potential protective effects of melatonin against 1,2-DCE-induced reproductive toxicity *in vivo*. (B, C) Melatonin significantly mitigated 1,2-DCE-induced atrophy of the germinal epithelium in mouse testicular sections, concurrent with a reduction in sperm abnormalities ($n = 5$). (D, E) The protective effect of melatonin is primarily exerted by inhibiting the IRF7-mediated pyroptosis pathway. We applied a Student's *t*-test to assess the statistical significance of the differences between the two groups ($n = 5$). Comparisons were made with ANOVA, followed by a Tukey post hoc multiple comparisons test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, vs. the control group or the indicated group.

Luo et al., 2023; Zi et al., 2022). Building upon these findings, our experimental data corroborated the lower incidence of abnormal spermatozoa observed *in vivo* in the melatonin-treated group following exposure to 1,2-DCE. Importantly, our study provided novel insights by demonstrating a significant reduction in pyroptosis signals in the testes of male mice exposed to 1,2-DCE when treated with melatonin. Pyroptosis is a form of programmed cell death that can lead to tissue damage (Yu et al., 2021). The ability of melatonin to attenuate pyroptosis signals suggests its potential in protecting testicular cells from 1,2-DCE-induced testicular toxicity. To unravel the underlying mechanisms through which melatonin exerts its protective effects against the elevated levels of piR-mmu-1019957 observed in spermatogenic cells induced by 1,2-DCE, we examined the expression of piR-mmu-1019957 and its downstream signaling events in 1,2-DCE-exposed GC-2 spd cells treated with melatonin. Our results revealed that melatonin treatment led to a reduction in IRF7 expression by inhibiting piR-mmu-1019957. Furthermore, the inhibition of IRF7 resulted in the suppression of downstream pyroptosis signaling events in 1,2-DCE-exposed GC-2 spd cells treated with melatonin. Consistent with these observations, the treatment with melatonin demonstrated a substantial decrease in the release of LDH compared to the group exposed to 1,2-DCE alone. LDH serves as a reliable indicator of cellular damage and its reduction further supports the protective effects of melatonin against 1,2-DCE-induced testicular toxicity (Pang et al., 2022). These findings support the notion that melatonin modulates piR-mmu-1019957-mediated regulation of IRF7 and subsequently mitigates pyroptosis in the context of 1,2-DCE-induced testicular toxicity. Additionally, previous evidence has illuminated that melatonin can regulate noncoding RNAs to alleviate pyroptosis (Liu et al., 2022a,b; Miao et al., 2022; Zhang et al., 2018c). Therefore, we deduced that piR-mmu-1019957 might be a novel molecular target for melatonin. In summary, our study provides compelling evidence for the beneficial effects of melatonin in mitigating 1,2-DCE-induced testicular damage. These results further support the protective effects of melatonin on testicular cells, highlighting its potential as a therapeutic intervention against 1,2-DCE-induced testicular toxicity.

The focus of this study is hazard identification and epigenetic mechanism exploration. Our *in vitro* experiments followed the Organization for Economic Cooperation and Development (OECD) guidelines for appropriate cytotoxicity and cell number. Therefore, we treated the GC-2 spd cells with 1,2-DCE at concentrations of 0, 7.5, 15, and 30 mM, with the cell viability of the 30 mM group being 77 % of the control group. We observed that the concentration of 1,2-DCE in the medium decreased after 24 h of treatment, likely due to volatility, leading to low recovery of 1,2-DCE after 24 h of exposure in the culture wells. For our selected endpoint, which is acute loss of cell viability or basal cytotoxicity, we considered the exposure concentration at the onset of exposure (C_{0h}) as the best reflection of the concentration causing the effect. This approach allowed us to mimic the toxic effects of low concentrations *in vivo* over a short period. A previous study (Withey and Karpinski, 1985) showed that the 1,2-DCE concentration in the blood was 300 mg/L (3 mM) 5–6 h after inhalation of 3000 ppm (666.7 mg/m³) 1,2-DCE in rats. The exposure levels in our *ex vivo* studies were comparable to the blood concentrations after inhalation of 1,2-DCE in existing animal studies. Furthermore, considering the reported cases published in the Chinese literature from 2006 to 2015, where the average concentration of 1,2-DCE in factories was 272.2 mg/m³ (range 7.4–1199 mg/m³) (Jiang et al., 2015), the 1,2-DCE doses applied in our *ex vivo* studies can be considered realistic.

5. Conclusion

Our study revealed that piR-mmu-1019957 plays a crucial role in positively regulating IRF7 to mediate 1,2-DCE-induced pyroptosis in mouse spermatocytes through the RIG-I-like cell signaling pathway. Furthermore, we demonstrated that melatonin effectively modulates the piR-mmu-1019957-mediated regulation of IRF7, resulting in the

mitigation of pyroptosis in the context of 1,2-DCE-induced toxicity. These findings have significant implications, suggesting that melatonin could be a practical therapeutic agent for attenuating male reproductive toxicity induced by 1,2-DCE in clinical practice.

CRediT authorship contribution statement

Bingli Zhang: Data curation. **Yizhou Zhong:** Conceptualization, Formal analysis, Visualization, Writing – original draft, Methodology. **Jiaxin Du:** Conceptualization, Formal analysis, Methodology, Visualization, Writing – original draft. **Rongyi Ye:** Data curation. **Bingchi Fan:** Data curation. **Yanhong Deng:** Data curation. **Ruobing Bai:** Data curation. **Yu Feng:** Data curation. **Xiaohong Yang:** Data curation. **Yuji Huang:** Data curation. **Boxuan Liang:** Data curation. **Jiewei Zheng:** Data curation. **Weifeng Rong:** Data curation. **Xingfen Yang:** Funding acquisition. **Zhenlie Huang:** Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2024.108480>.

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